# Three Experiments for Rigorous University Replication

A concept note on testing extraordinary claims about consciousness under clean, ethical, preregistered, and publishable conditions.

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| **Purpose** | Rough proposals and indicative budgets to support early conversations with university partners. |
| **Status** | Draft for discussion — figures are order-of-magnitude estimates. |

**Framing.** All three studies are designed as *replication / falsification* studies. The aim is not to prove extraordinary claims, but to test them under conditions that are publishable even when the result is negative. A clean null result is a valuable scientific contribution; a positive result would be highly significant.

**Costing basis.** Budgets are grounded in **Cuenca, Ecuador** conditions: local academic and clinical salaries, and the assumption that major equipment (EEG, fMRI, infusion pumps, monitoring) is **borrowed from partner institutions rather than purchased**. Equipment lines therefore cover access, transport, calibration, and consumables only. All figures are in USD (Ecuador's currency).

**Two scenarios.** Each study is presented as a range. The upper bound is the **fully-costed** figure at market rates. The lower bound is the **partnership-leveraged** figure, which assumes maximal in-kind support: donated equipment and scanner time, clinical and academic collaborators contributing as co-investigators, and graduate researchers embedding the work within their own theses. The realistic operating budget typically sits between the two, since skilled personnel time is the least compressible cost.

## Executive Summary

This note outlines three candidate experiments that could anchor a university research agenda on altered states of consciousness, neurophenomenology, and the scientific study of extraordinary human experience. Each is summarised below with a core idea, a proposed design, required infrastructure, ethical considerations, and an indicative budget. A comparative budget table and a longer pipeline of ten further candidates are included at the end.

| Study | Core question | Indicative budget | Risk / complexity |
|---|---|---|---|
| **1. Extended-State DMT (DMTx)** | Can the DMT state be safely extended and characterised in real time? | $40k – $150k | High — controlled substance, medical |
| **2. Brain-to-Brain Correlation** | Do separated, paired subjects show correlated EEG responses? | $17k – $55k | Medium — design rigor is everything |
| **3. Trance / Healer Neurophysiology** | Do practitioners enter a reproducible, distinct brain state? | $18k – $50k | Low–Medium — recruitment-limited |

**Recommended sequencing.** Studies 2 and 3 are lower-cost, lower-risk, and faster to ethics approval; either could serve as a credible first project that demonstrates rigor to partners and funders. Study 1 (DMTx) is the scientific flagship but carries the heaviest regulatory, medical, and budgetary load and is best pursued once institutional relationships and a clinical partner are in place.

EXPERIMENT 1

# Extended-State DMT / DMTx

## Core idea

Rather than studying DMT as a very short (5–15 minute) experience, this protocol uses a controlled intravenous loading dose followed by a target-controlled maintenance infusion to **extend and stabilise the DMT state** for an experimentally useful window. This makes it possible to study the phenomenology, neural correlates, and any recurring structures of the experience while collecting real-time physiological and neuroimaging data. The informal DMT / laser-perception idea we discussed would sit inside this framework as a small, blinded perceptual sub-study rather than as the flagship aim.

## Proposed design

- **Design:** open-label pharmacokinetic / pharmacodynamic pilot, n ≈ 8–12 healthy, psychedelic-experienced volunteers, within-subject dose escalation.
- **Primary outcomes:** safety and tolerability of extended infusion; feasibility of maintaining a stable subjective state at a target plasma level.
- **Secondary outcomes:** EEG signatures (spectral power, complexity / entropy, connectivity); validated phenomenology scales; structured post-session interviews.
- **Optional blinded sub-study:** randomised, blinded perceptual tasks delivered during the stable state to test specific claims under controlled conditions.
- **Analysis:** preregistered, with predefined safety stopping rules and primary endpoints.

| Key infrastructure & equipment | Ethical & safety considerations |
|---|---|
| • Clinical research unit with anaesthesia-grade monitoring (partner hospital)<br>• Target-controlled IV infusion pump(s) — borrowed<br>• Research-grade EEG & physiological monitoring — borrowed<br>• Pharmaceutical-grade DMT and hospital pharmacy handling<br>• Controlled-substance storage and chain-of-custody | • Schedule I / controlled-substance licensing and import permits<br>• Full IRB / research-ethics board review<br>• On-site physician and resuscitation capability<br>• Rigorous screening; cardiac and psychiatric exclusions<br>• Trained psychological support and integration |

## Indicative timeline

Approximately 18–24 months: 6–9 months for regulatory, ethics, and drug supply; 3 months setup and staff training; 6–9 months data collection; 3–6 months analysis and write-up.

## Indicative budget (rough / general)

| Cost category | Estimate (USD) |
|---|---|
| Regulatory, ethics & controlled-substance permits (Ecuador) | $8,000 |
| Pharmaceutical-grade DMT supply & secure handling | $20,000 |

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|---|---|
| Clinical & medical staff (physician, anaesthesia, nurse, psychiatrist) — local rates | $25,000 |
| Borrowed EEG / monitoring — access, transport, calibration & consumables | $8,000 |
| Core research personnel (PI, postdoc, RAs, analyst — ~1 yr, Cuenca rates) | $40,000 |
| Participant screening, compensation & facility | $8,000 |
| Data analysis, computing, dissemination & contingency | $6,000 |
| **Indicative total** | **≈ $90,000 – $150,000 (fully costed)** |

*With maximal in-kind institutional support — donated drug synthesis, clinical collaborators contributing as co-investigators, borrowed instrumentation, and graduate researchers embedding the work within their theses — the partnership-leveraged floor falls to approximately $40,000.*

## Key references & links


- Gallimore & Strassman (2016), Frontiers in Pharmacology
- Extended DMT study (PMC10851633)
- Continuous DMT infusion study (PMC12032411)

EXPERIMENT 2

# Brain-to-Brain Correlation / “Transferred Potential” Replication

## Core idea

Inspired by Jacobo Grinberg’s controversial work on “transferred potentials,” in which two people are placed in separate rooms, one receives a stimulus, and the other’s brain activity is monitored for correlated responses. The claim is extraordinary — which is exactly why it deserves a **modern, rigorously controlled replication**. The scientific value here lies almost entirely in design quality: double-blinding, shielding, randomised timing, and fully automated, preregistered analysis remove the loopholes that made the original work impossible to interpret.

## Proposed design

- **Design:** paired-subject EEG study, ≈ 25–30 pairs (emotionally bonded and stranger pairs), “sender” and “receiver” in separate electromagnetically shielded rooms.
- **Stimulus:** randomised, computer-triggered sensory stimuli to the sender at times unknown to the receiver and to all experimenters present.
- **Primary outcome:** preregistered test for time-locked correlated activity in the receiver’s EEG during sender-stimulation vs. matched null windows.
- **Controls:** sham/no-sender blocks, sensor-level artifact rejection, and analyst blinding to condition labels.
- **Statistics:** preregistered effect-size threshold, Bayesian + frequentist criteria; a well-powered null is a publishable outcome.

**Key infrastructure & equipment**
- Two research-grade EEG systems, time-synchronised — borrowed
- Electromagnetically shielded / Faraday-isolated rooms (existing facility)
- Hardware trigger & precision timing infrastructure
- Automated, preregistered analysis pipeline
- Audio/RF isolation verification

**Ethical & safety considerations**
- IRB / ethics approval (low physical risk)
- Informed consent; right to withdraw
- Pre-registration on OSF / AsPredicted before data collection
- Independent verification of room isolation
- Open data and analysis code for full transparency

## Indicative timeline

Approximately 12–15 months: 2–3 months ethics and preregistration; 2 months setup and isolation testing; 5–6 months paired data collection; 3–4 months analysis and write-up.

## Indicative budget (rough / general)

| Cost category | Estimate (USD) |
|---|---|
| Borrowed EEG — transport, caps/electrodes, gel, calibration | $4,000 |
| Shielded-room setup & isolation testing (existing facility) | $5,000 |

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|---|---|
| Core personnel (PI part-time, postdoc, 2 RAs, analyst — ~1 yr, Cuenca rates) | $30,000 |
| Participant recruitment & compensation (~60 participants) | $4,000 |
| Preregistration, replication design & statistical consulting | $3,000 |
| Computing, open-data hosting & dissemination | $2,000 |
| **Indicative total** | **≈ $35,000 – $55,000 (fully costed)** |

*With borrowed instrumentation, free use of an existing shielded facility, and student researchers carrying the data collection, the partnership-leveraged floor falls to approximately $17,000.*

## Key references & links

- Jacobo Grinberg — background & publication reference (Wikipedia, ES)
- Related separated-subject EEG study (PubMed 12972348)

EXPERIMENT 3

# Shamans / Healers / Trance Practitioners under EEG or fMRI

## Core idea

Study shamans, healers, meditators, or trance practitioners while they enter their claimed healing or altered state, and compare their brain activity and physiology against matched controls and against their own ordinary rest, imagination, and meditation. Crucially, this study does **not** need to begin by claiming that "healing energy" is real. The first, stronger scientific question is simply: *do trained practitioners enter a reproducible neurophysiological state that is measurably different from ordinary baselines?* If yes, a later phase can test whether that state has any measurable effect on recipients.

## Proposed design

- **Design:** within-subject, multi-condition comparison (trance vs. rest vs. imagery vs. meditation), ≈ 15–25 experienced practitioners + matched controls.
- **Primary outcome:** reproducible, practitioner-specific EEG/fMRI signature that discriminates the trance state from control conditions above chance.
- **Secondary outcomes:** autonomic measures (HRV, electrodermal activity, respiration); state classification with cross-validated machine learning.
- **Phase 2 (optional):** blinded recipient study testing for any physiological or perceptual effect on a second person.
- **Analysis:** preregistered classification accuracy and effect-size thresholds.

| Key infrastructure & equipment | Ethical & safety considerations |
|---|---|
| • Research-grade EEG and/or fMRI scanner access — borrowed<br>• Autonomic / peripheral physiology recording (HRV, EDA) — borrowed<br>• Synchronised audio-video for state annotation<br>• Analysis workstation & ML pipeline<br>• Quiet, controlled recording environment | • IRB / ethics approval; culturally respectful protocol<br>• Community consultation & fair compensation for practitioners<br>• Informed consent honouring practitioners' traditions<br>• Clear non-medical-claim framing in all materials<br>• Pre-registration of hypotheses and analysis |

## Indicative timeline

Approximately 12–18 months: 2–3 months ethics, community engagement, and recruitment; 5–7 months data collection (recruitment-paced); 3–5 months analysis and write-up.

## Indicative budget (rough / general)

| Cost category | Estimate (USD) |
|---|---|
| Borrowed EEG / fMRI — scanner access fees & consumables (~40 sessions) | $5,000 |
| Core personnel (postdoc, RAs, data analyst — ~1 yr, Cuenca rates) | $28,000 |

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| Practitioner recruitment, travel & compensation | $6,000 |
| Peripheral physiology (HRV, EDA) — borrowed / low-cost | $2,000 |
| Ethics, preregistration, dissemination & contingency | $3,000 |
| **Indicative total** | **≈ $30,000 – $50,000 (fully costed)** |

*With donated scanner / EEG time, borrowed physiology hardware, and student researchers leading data collection, the partnership-leveraged floor falls to approximately $18,000.*

## Key references & links

- Cognitive / shamanic-style trance research (arXiv 2509.19254)
- Trance & neurophysiology — general background (Wikipedia)

## Comparative Budget Overview

The figures below are **rough, order-of-magnitude estimates** for pilot studies run in **Cuenca, Ecuador**, using **local salaries** and **borrowed equipment** from partner institutions (no major hardware purchases). Each study is shown across two scenarios: a **partnership-leveraged** figure that assumes maximal in-kind support, and a **fully-costed** figure at market rate. The largest variable in both is skilled personnel time.

| Study | Partnership-leveraged | Fully costed | Duration | Role |
|---|---|---|---|---|
| **1 · Extended-State DMT (DMTx)** | ~$40k | $150k | 18–24 months | Flagship — later phase |
| **2 · Brain-to-Brain Correlation** | ~$17k | $55k | 12–15 months | Strong candidate |
| **3 · Trance / Healer Neurophysiology** | ~$18k | $50k | 12–18 months | Strong candidate |
| **Combined programme (all three)** | **~$75k** | **$255k** | ~2–3 years phased | — |

The *partnership-leveraged* column reflects maximal in-kind support (donated equipment and scanner time, collaborators contributing as co-investigators, student-led data collection); the *fully-costed* column prices every line at market rate. A realistic operating budget usually lands between the two, since skilled personnel time is the least compressible cost. Studies 2 and 3 are modest enough to run as graduate / faculty research projects; Study 1 (DMTx) carries the clinical, drug, and regulatory load that keeps it the most expensive even at local prices.

## Cross-Cutting Principles

**Replication, not advocacy.** Every study is framed to be publishable whether the result is positive or null. We are testing claims, not defending them.

**Preregistration by default.** Hypotheses, sample sizes, and analysis plans are registered before data collection on OSF or a comparable platform, protecting against bias and post-hoc storytelling.

**Ethics first.** Each study is designed to be ethically approvable at a mainstream university, with appropriate review, consent, safety, and (for Study 3) cultural respect.

**Open and transparent.** Open data and analysis code wherever ethically possible, so results can be independently verified — the single most important safeguard for controversial questions.

## Broader Pipeline — Ten Further Candidates

Beyond the three priority studies, the following are strong additional replication candidates in consciousness and altered-states research, available to build out a longer programme:

1. Real-time communication with lucid dreamers during REM sleep
2. Psilocybin and increased brain-state repertoire
3. Ayahuasca and brain entropy / network reorganization
4. Expert meditators and high-amplitude gamma synchrony
5. Hypnosis and changes in pain, agency, and self-perception
6. Breathwork-induced non-ordinary states without psychedelics
7. Ganzfeld / sensory deprivation and internally generated perception
8. Near-death experience research and end-of-life brain activity
9. Anesthesia and perturbational complexity as a measure of consciousness
10. Virtual-reality body-transfer or out-of-body illusion experiments

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Prepared by Ney Torres · 22 June 2026 · Draft concept note for discussion with university partners. All budget figures are indicative and subject to refinement with a host institution.